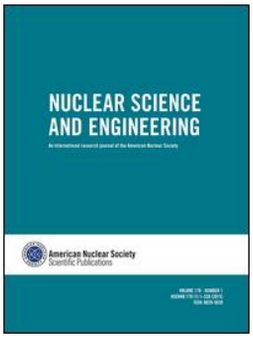

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Stochastic Processes in a Subcritical Nuclear Reactor in the Presence of a Fission Source

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Abstract — *The forward Kolgomorov equation is written for the case of a subcritical reactor monitored by two detectors and excited by a fission source located inside a fission chamber (an arrangement currently in use to measure reactivities). The marginal distribution of neutrons is shown to be given by the negative binomial distribution with an amplified correlation as compared to the case of a photoneutron source. The amplification allows the definition of an equivalent factor D_{eq} for the Diven factor, which makes possible the application of formulas originally derived for interpretation of noise measurement in the presence of a photoneutron source to the case of a fission source. The ratio of the correlations measured under the successive presence of both kind of sources allows the direct measurement of the effective delayed fraction, β_{ef}. The factor D_{eq} is proven to be consistent with a derivation based on the Schottky prescription for the noise source.*

Processus stochastiques intervenant dans un réacteur nucléaire sous-critique en présence d'une source de fission

Résumé — *L'équation prospective de Kolmogoroff est écrite pour le cas d'un réacteur sous-critique surveillé par deux détecteurs et excité par une source de fission placée dans une chambre de fission (une configuration couramment utilisée pour mesurer les réactivités). On montre que la fonction de distribution marginale de neutrons est donnée par la distribution négative binomiale avec une corrélation augmentée par rapport au cas d'une source de photoneutrons. L'augmentation permet de définir un facteur D_{eq} équivalent au facteur Diven de sorte que les formules initialement dérivées pour pouvoir interpréter le bruit mesuré en présence d'une source de photoneutrons s'appliquent au cas d'une source de fission. Le rapport des corrélations mesurées en présence successive des deux types de sources permet de mesurer directement la fraction effective retardée β_{ef}. On prouve que D_{eq} conforme à une dérivation s'appuyant sur la prescription de Schottky relative à la source du bruit.*

Stochastische Prozesse in einem unterkritischen Kernreaktor mit Spaltquelle

Zusammenfassung — *Die Vorwärtsgleichung von Kolgomorov wird für den Fall eines mit zwei Detektoren überwachten und mit einer in einer Spaltkammer angebrachten Spaltquelle (diese Anordnung ist für Rektivitätsmessungen üblich) angeregten unterkritischen Reaktors aufgeschrieben. Es wird gezeigt, daß die Marginalverteilung der Neutronen durch die negative Binomialverteilung mit einer im Vergleich zum Fall mit Photoneutronenquelle verstärkten Korrelation gegeben ist. Durch die Verstärkung ist es möglich, einen dem Diven-Faktor äquivalenten Faktor D_{eq} zu definieren, so daß Formeln, die ursprünglich für die Interpretation des in Gegenwart einer Photoneutronenquelle gemessenen Rauschens abgeleitet wurden, auf den Fall angewandt werden können, in dem eine Spaltquelle vorliegt. Das Verhältnis der gemessenen Korrelationen, bei denen beide Arten von Quellen nacheinander vorlagen, ermöglicht die direkte Messung des effektiven Anteils verzögerter Neutronen β_{ef}. Es wird nachgewiesen, daß D_{eq} mit einer Ableitung auf der Grundlage der Schottkyschen Vorschrift für die Rauschquelle übereinstimmt.*

I. INTRODUCTION

Theoretical *ad hoc* models developed for 19 different reactor noise measurement techniques during the last 40 years were unified in a single theory by Pacilio et al.[1] According to this theory, any noise technique can be described as an operator acting on the probability generating function F, which connects such totally different experimental techniques as

1. probability of n counts in a sampling time Δt
2. moments of the distribution of counts in Δt
3. correlation in time domain
4. correlation of polarities
5. correlation in frequency domain.

Pacilio et al. calculated F for the case of neutron sources that produce only one neutron per generating event (like photoneutron sources). Their formalism does not include, however, the ^{252}Cf source method developed by Mihalczo et al.[2]; recently, we generalized the formalism to include this new noise technique.[3] A very detailed theory has already been developed[4] to include the space and energy variables in equations describing the stochastic field for both types of sources. The complications of this theory, however, limit its practical applications to the calculation of the first two moments of the distribution of counts instead of the calculation of the probability generating function. Therefore, it is useful to explore the possibility of obtaining the entire probability generating function using a point reactor model. The forward Kolgomorov equation for F is written for the case of the ^{252}Cf source[a] method and is partially solved in terms of the marginal distribution of neutrons. An equivalent factor for the Diven factor is found for both the marginal distribution of neutrons and the reduced variance of the distribution of counts. This result is proven to be consistent with a derivation based on the Schottky noise source.

II. THE KOLGOMOROV EQUATION FOR THE ^{252}Cf SOURCE METHOD

Figure 1 shows a schematic arrangement used to measure reactivities with the ^{252}Cf source method. The source is within a fission chamber (labeled "detector 1") that essentially measures the rate of spontaneous fissions in the ^{252}Cf deposit; detectors 2 and 3 are neutron detectors. The ratio of cross-power spectral densities between different pairs of detectors, $G_{12}^{*}G_{13}/G_{11}G_{23}$, was related[2] to the reactivity of the system

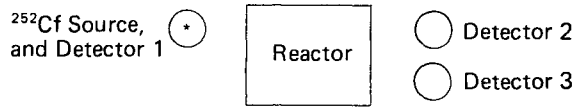

Fig. 1. Schematic arrangement for the ^{252}Cf source noise method to measure reactivities. The source is within detector 1.

measured in absolute units. The relationship was obtained in Ref. 2 by using a Langevin/Schottky model.

Let us call S (s^{-1}) the fission rate in the ^{252}Cf and $P(n,r_1,r_2,r_3,t)$ the probability of having at time t n neutrons in the reactor and r_1, r_2, and r_3 counts in detectors 1, 2, and 3 of efficiencies ϵ_i ($i=1,2,3$), respectively. The values Λ_c, Λ_f, Λ_{d_2}, and Λ_{d_3} (s^{-1}) denote the probabilities per unit time and per neutron for the capture and fission in the reactor, and for the captures in detectors 2 and 3, respectively.

The equation for the detailed balance of probabilities is written below with the convention that an independent variable is only written when it is different from either n, r_1, r_2, or r_3:

$$\begin{aligned}P(n,r_1r_2,r_3,t+\Delta t) &= \sum_m P(n-m,t)S\Delta t p_s(m)(1-\epsilon_1)\\ &+ \sum_m P(n-m,r_1-1,t)S\Delta t p_s(m)\epsilon_1\\ &+ (n+1)P(n+1,t)\Lambda_c\Delta t\\ &+ \sum_j P(n-j+1,t)\Lambda_f(n-j+1)p_F(j)\Delta t\\ &+ (n+1)P(n+1,r_2-1,t)\Lambda_{d_2}\epsilon_2\Delta t\\ &+ (n+1)P(n+1,t)\Lambda_{d_2}(1-\epsilon_2)\Delta t\\ &+ (n+1)P(n+1,r_3-1,t)\Lambda_{d_3}\epsilon_3\Delta t\\ &+ (n+1)P(n+1,t)\Lambda_{d_3}(1-\epsilon_3)\Delta t\\ &+ \{1-[S+n(\Lambda_c+\Lambda_f+\Lambda_{d_2}+\Lambda_{d_3})]\Delta t\}P(t)\ ,\end{aligned} \tag{1}$$

where $p_s(m)$ and $p_F(j)$ are the probabilities of having m and j neutrons after a fission in the ^{252}Cf or in the fuel, respectively.

Equation (1) includes the probability of having an extra count in detector 1 after the spontaneous fission in the ^{252}Cf [second term in the right side of Eq. (1)]. It also includes explicitly the efficiencies ϵ_i. The remaining terms are similar to those discussed in Ref. 5.

Defining the probability generating functions,

$$F(x,z_1,z_2,z_3,t) = \sum_{n,r_1,r_2,r_3} P(n,r_1,r_2,r_3,t)x^n z_1^{r_1}z_2^{r_2}z_3^{r_3}\ , \tag{2}$$

[a]In Ref. 5, the forward Kolgomorov equation was written for the general case of variable sources without including the detector inside where the source is located.

$$f_s(x) = \sum_m p_s(m) x^m \ , \tag{3}$$

and

$$f_F(x) = \sum_j p_s(j) x^j \ , \tag{4}$$

the system of differential equations [Eq. (1)] can be transformed into a single differential equation:

$$\frac{\partial F}{\partial t} = S\{f_s(x)[1+\epsilon_1(z_1-1)]-1\}F + \{(\Lambda_c+\Lambda_{d_2}+\Lambda_{d_3})(1-x)+\Lambda_f[f_F(x)-x] + \epsilon_2\Lambda_{d_2}(z_2-1)+\epsilon_3\Lambda_{d_3}(z_3-1)\}\frac{\partial F}{\partial x} \ . \tag{5}$$

Following the approximation used and discussed in Ref. 1, $f_s(x)$ and $f_F(x)$ are expanded up to the second order in $(x-1)$ so that Eq. (5) becomes the following more manageable equation:

$$\frac{\partial F}{\partial t} = S(x-x_1)(x-x_2)[1+\epsilon_1(z_1-1)]\frac{\overline{\nu_s(\nu_s-1)}}{2}F + \alpha Y_1(x-x_3)(x-x_4)\frac{\partial F}{\partial x} \ . \tag{6}$$

In Eq. (6), ν_s is the number of neutrons per fission in the ^{252}Cf (the overbar indicates average), α (s^{-1}) is the prompt neutron decay constant, and

$$x_{\substack{1\\2}} = 1 - \frac{1}{\bar{\nu}_s D_s} \pm \left[\frac{1}{(\bar{\nu}_s D_s)^2} - \frac{2}{(\bar{\nu}_s)^2 D_s} \times \frac{\epsilon_1(z_1-1)}{1+\epsilon_1(z_1-1)}\right]^{1/2} \ , \tag{7}$$

$$x_{\substack{3\\4}} = 1 + \frac{1 \pm Z}{2Y_1} \ , \tag{8}$$

$$Z = \left[1 + 2\sum_{i=1}^{3} Y_i^*(1-z_1)\right]^{1/2} \ , \tag{9}$$

$$Y_i^* = \frac{2Y_1\Lambda_{d_i}\epsilon_i}{\alpha} \ , \tag{10}$$

and

$$Y_1 = \frac{\Lambda_f D_F}{2\alpha}(\bar{\nu}_F)^2 \ . \tag{11}$$

In the above equations, ν_F is the number of neutrons per fission in the fuel, and D_S and D_F are the Diven factors for the source and the fuel, respectively.

We seek a solution of Eq. (6) under the following circumstances:

1. The reactor is in equilibrium with the source S; i.e., the marginal distribution of neutrons is independent of time.
2. The detectors are activated at time $t=0$; i.e., $\epsilon_i = 0$ $(i=1,2,3)$ for $t \le 0$, $\epsilon_i \ne 0$ for $t > 0$.

Mathematically,

$$F(x,z_1,z_2,z_3,t=0) = K(x) \tag{12}$$

and

$$F(x,z_1=1,z_2=1,z_3=1,t) = K(x) \ . \tag{13}$$

Equation (12) states the initial condition for F since the detectors are activated at $t=0$ ($r_1=r_2=r_3=0$ for $t \le 0$). Equation (13) states the ergodic hypothesis; i.e., at steady state the distribution of neutrons is independent on time.

The value $K(x)$ is the final solution $A(x,t' \to \infty)$ of the Kolgomorov Eq. (6), with the detectors deactivated ($\epsilon_i = 0$; $i=1,2,3$):

$$\frac{\partial A}{\partial t'} = S\frac{\overline{\nu_s(\nu_s-1)}}{2}(x-1)(x-y_0)A + \alpha Y_1(x-1)(x-y_1)\frac{\partial A}{\partial x} \ , \tag{14}$$

where

$$y_0 = 1 - \frac{2}{\bar{\nu}_s D_s} \tag{15}$$

and

$$y_1 = 1 + \frac{1}{Y_1} \ . \tag{16}$$

Equation (14) describes the approach to equilibrium when the source is switched on at time $t'=0$ long before the detectors are activated.

III. THE MARGINAL DISTRIBUTION OF NEUTRONS

In this section, we solve Eq. (14) to observe, from the stochastic point of view, how the system approaches steady condition. The asymptotic solution is the initial condition for F; it is also the marginal distribution of neutrons for all the time.

Elementary solutions of Eq. (14) can be found by the method of separation of variables. The following product $X(x)T(t')$ is one of these solutions:

$$X(x)T(t') = \frac{\exp[-\lambda q(x)]\exp(-\lambda t')}{H(x)} \ , \tag{17}$$

where

$$H(x) = (x-y_1)^{S_n f/Y_1\alpha}\exp[S_n(f-1)x/\alpha] \tag{18}$$

$$S_n = \bar{\nu}_s S \tag{19}$$

$$f = 1 + \frac{\bar{\nu}_s D_s}{2Y_1} \tag{20}$$

$$q(x) = \frac{1}{\alpha} \ln \frac{x - y_1}{x - 1} \ . \tag{21}$$

In these equations, S_n is the neutron source and λ is the separation constant.

The general solution for A is

$$A(x,t') = \frac{1}{H(x)} \int_0^\infty C(\lambda) \exp[-\lambda(q + t')] \, d\lambda \ , \tag{22}$$

where $C(\lambda)$ is determined by the initial condition on A; $Q(x) = A(x,t' = 0)$, i.e., by

$$\int_0^\infty C(\lambda) \exp(-\lambda q) \, d\lambda = H(x) Q(x) \equiv B(q) \ . \tag{23}$$

The value $B(q)$ can be found by substituting $x = x(q)$ [given by the inverse of Eq. (21)] in the definition of B; A is given then by

$$A(x,t') = \frac{B(q + t')}{H(x)} \ . \tag{24}$$

Instead of solving the functional form of B, one can use its definition. Thus,

$$A(x,t') = \frac{H[x(q + t')] Q[x(q + t')]}{H(x)} \ , \tag{25}$$

where from Eq. (21)

$$x(q + t') = 1 + \frac{1}{Y_1 \left[1 - \exp(\alpha t') + \dfrac{\exp(\alpha t')}{Y_1(x - 1)} \right]} \ . \tag{26}$$

In our case, we assume that the source is switched on at $t' = 0$; consequently, there are no neutrons in the system at time $t' \leq 0$ and $Q = 1$. Therefore, A is given by

$$\begin{aligned} \ln A(x,t') &= - \frac{S_n f}{Y_1 \alpha} \ln\{1 + Y_1(1 - x)[1 - \exp(-\alpha t')]\} \\ &\quad + \frac{S_n}{\alpha} (f - 1)(1 - x) \\ &\quad \times \left\{ 1 - \frac{\exp(-\alpha t')}{1 + Y_1(1 - x)[1 - \exp(-\alpha t')]} \right\} \ . \end{aligned} \tag{27}$$

The approach to the steady-state distribution is determined by the prompt decay constant α. After a few neutron lifetimes, the marginal distribution of neutrons is given by

$$\begin{aligned} \ln K(x) &= - \frac{S_n f}{Y_1 \alpha} \ln[1 + Y_1(1 - x)] \\ &\quad + \frac{S_n}{\alpha} (f - 1)(1 - x) \ . \end{aligned} \tag{28}$$

Equation (28) is equivalent to the negative binomial or Polya distribution

$$K(x) = [1 + Y_1(1 - x) f]^{-\bar{n}/f Y_1} \tag{29}$$

up to the second order in $(1 - x)$. The departure of the reduced variance from the Poisson statistic is

$$\psi_n \equiv \frac{\sigma^2(n)}{\bar{n}} - 1 \equiv \frac{\overline{n^2} - (\bar{n})^2}{\bar{n}} - 1 = f Y_1 \ . \tag{30}$$

Equations (29) and (30) are also valid in the case of photoneutron sources (factor f becomes equal to unity). In our case, $f > 1$; the extra degree of randomness due to the nature of the ^{252}Cf source produces a neutron field that is noisier than usual. Now not only the generating event is at random (the fission in the ^{252}Cf), but also the number of neutrons per generating event.

Equations (29) and (30) become formally identical to the case of photoneutron sources if we define an equivalent factor for the Diven factor of the fuel

$$D_{eq} \equiv f D_F \ . \tag{31}$$

The gain factor f from Eq. (20) can be written in terms of the effective neutron delayed fraction β_{ef} and the reactivity in dollar units \$:

$$f = 1 + \frac{\bar{\nu}_s D_s}{\bar{\nu}_F D_F} \beta_{ef}(1 + |\$|) \ . \tag{32}$$

IV. CALCULATION OF THE REDUCED VARIANCE OF THE NUMBER OF COUNTS

In Sec. III, we demonstrated that the neutron field in the presence of the ^{252}Cf source obeys formally the same type of statistics as the one in the presence of a photoneutron source. But the neutron field is not a direct observable. We observe the field through its interaction with the detectors; therefore, it is important to know whether the same generalization can be extended to the statistic of the counters. As discussed in Ref. 1, all neutron noise techniques are interrelated; consequently, it would be sufficient to study one technique, for example, the reduced variance or Feynman et al. method.[6]

The Feynman et al. method is based on the measurement of the departure of the reduced variance of the number of counts in one detector from the Poisson statistics. This departure is defined by

$$\psi \equiv \frac{\sigma^2(r)}{\bar{r}} - 1 \equiv \frac{\overline{r^2} - (\bar{r})^2}{\bar{r}} - 1 \ , \tag{33}$$

where r is the number of counts accumulated between time 0 and time t.

In the case of one detector, Eq. (5) is reduced to

$$\frac{\partial F}{\partial t} = S[f_s(x) - 1]F + \{(\Lambda_c + \Lambda_d)(1 - x) + \Lambda_f[f_F(x) - x] + \epsilon\Lambda_d(z - 1)\}\frac{\partial F}{\partial x} , \quad (34)$$

where the subscript on ϵ, Λ_d, z, and r is omitted. To calculate $\bar{r}$ and $\overline{r^2}$, partial derivatives of Eq. (34) with respect to x and z are evaluated at $x = z = 1$.

The equations for the first and second moments of the distribution are then

$$\frac{d\bar{n}}{dt} = S_n - \alpha\bar{n} , \quad (35)$$

$$\frac{d\bar{r}}{dt} = \epsilon\Lambda_d\bar{n} , \quad (36)$$

$$\frac{d\overline{nr}}{dt} = S_n\bar{r} - \alpha\overline{nr} + \epsilon\Lambda_d\overline{n(n-1)} , \quad (37)$$

$$\frac{d}{dt}\overline{n(n-1)} = S\overline{\nu_s(\nu_s - 1)} + 2S_n\bar{n} + \Lambda_f\overline{\nu_F(\nu_F - 1)n} - 2\alpha\overline{n(n-1)} , \quad (38)$$

and

$$\frac{d}{dt}\overline{r(r-1)} = 2\Lambda_d\epsilon\overline{nr} . \quad (39)$$

The first term on the right side of Eq. (38) would be zero in the case of a photoneutron source. The solutions of Eqs. (35) through (39) allow the calculation of $\psi_r(t)$:

$$\psi_r(t) = E\left[1 - \frac{1 - \exp(-\alpha t)}{\alpha t}\right] , \quad (40)$$

where

$$E = \frac{eD_F}{\beta_{ef}^2(1 + |\$|)^2}f , \quad (41)$$

where e is the customary efficiency of the detector with respect to the total fission rate and f is given by Eq. (32). If we adopt the definition of Eq. (31) for an equivalent factor for the Diven factor, Eqs. (40) and (41) become formally identical to the formulas applicable in the case of a photoneutron source. This result suggests that all noise techniques based on measurement of the second moment in the presence of the ^{252}Cf source can be described by the formulas derived for the case of a photoneutron source, provided we made the formal substitution of D_{eq} for D_F.

Factor f can be measured as the ratio between the correlations measured first in the presence of a ^{252}Cf source and then a photoneutron source; a fit of f as function of the reactivity [according to Eq. (32)] allows a direct measurement of β_{ef}.

V. DERIVATION OF THE EQUIVALENT FACTOR FOR THE DIVEN FACTOR ACCORDING TO THE SCHOTTKY FORMALISM

In this section, we show that the previous definition of the equivalent factor for the Diven factor can be also derived from the calculation of the source of noise. The Schottky[7] prescription of the source of noise G_{ss} is

$$G_{ss} = 2\sum_i q_i^2\bar{m}_i , \quad (42)$$

where i is a process occurring at the average rate $\bar{m}_i$ and producing a net number of neutrons q_i. Table I summarizes the processes corresponding to our case.

The source of noise can be written in terms of formulas for q_i and $\bar{m}_i$, given in Table I, as

$$G_{ss} = 2S_n\left(\frac{\overline{\nu_s^2}}{\bar{\nu}_s} + \frac{\bar{\nu}_F D_F - \rho}{\beta_{ef} - \rho}\right) , \quad (43)$$

where $\rho = (k_{ef} - 1)/k_{ef}$ is the reactivity in absolute units. Equation (43) becomes an equation formally identical to the case of a photoneutron source if we define an equivalent factor D'_{eq} for the Diven factor such that

$$2S_n\left(\frac{\overline{\nu_s^2}}{\bar{\nu}_s} + \frac{\bar{\nu}_F D_F - \rho}{\beta_{ef} - \rho}\right) \equiv 2S_n\left(1 + \frac{\bar{\nu}_F D'_{eq} - \rho}{\beta_{ef} - \rho}\right) . \quad (44)$$

The D'_{eq} derived from Eq. (44) is identical to D_{eq} given by Eqs. (31) and (32).

VI. CONCLUSIONS

The Kolgomorov equation was written for the case of a reactor excited with a fission source and monitored to measure the reactivity according to the method developed by Mihalczo et al.[2] The preliminary formalism, necessary to obtain the marginal distribution of counts, was completed. It consisted of the

TABLE I

Derivation of the Noise Source

Process	Rate ($\bar{m}_i$)	Net Number of Neutrons (q_i)
1[a]	$\bar{n}\Lambda_c$	-1
2[b]	$\bar{n}\Lambda_f P_F(\nu)$	$\nu - 1$
3[c]	$SP_S(\nu)$	ν

[a]Capture and leakage.
[b]Fission in the fuel.
[c]Spontaneous fission in the source.

calculation of the initial condition for the probability generating function. Due to ergodicity, the initial condition is also the marginal distribution of neutrons after transients introduced by the source have died out. The distribution up to the second-order moment is identical to the one in the presence of a photoneutron source except that the correlation term is amplified by the factor f given by Eq. (32). This factor allows the definition of an equivalent factor for the Diven factor. Thus, the formulas developed originally to interpret noise measurements in the presence of a photoneutron source may be translated to the case of a fission source.

The theoretical results suggest a new method to measure β_{ef}: The ratio between the correlations measured in the presence of both types of sources is linearly related to β_{ef} according to Eq. (32).

The same equivalent factor for the Diven factor is obtained from the calculation of Schottky's noise source, which implies consistency of the summarized results with a Langevin approach.

ACKNOWLEDGMENT

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